

Exp No: 17 UART INTERFACING USING INTERRUPT IN KEIL

PROGRAM:

```
#include <reg51.h>

void serial_ISR(void) interrupt 4
{
    if(RI)
    {
        P1 = SBUF;
        RI = 0;
    }
}

void main()
{
    TMOD=0x20;
    TH1=0xFD;
    SCON=0x50;
    TR1=1;
    ES=1;
    EA=1;
    while(1);
}
```

}

Disassembly

19: while(1):

C:0x080F 80FE Sjmp C:080F

C:0x0811 787F MOV R0,#0x7F

C:0x0813 E4 CLR A

exp 17.c

```
1#include <reg51.h>
2void serial_isr(void) interrupt 4
3{
4    if(RI)
5    {
6        P1 = SBUF;
7        RI = 0;
8    }
9}
10void main()
11{
12    TMOD=0x20;
13    TH1=0xFD;
14    SCON=0x50;
15    TR1=1;
16    ES=1;
17    EA=1;
18    while(1);
19}
```

Parallel Port 1

Post 1: 7 Bits 0

P1: 0xFF

Pin: 0xFF

Serial Channel

Mode: 8-Bit var Baudrate

SCON: 0x50 SBUF: 0x41

☐ SM2 ☒ REN

☐ TB8 ☐ RB8

Baudrate: 9600

☐ SMOD ☐ TI ☐ RI

Interrupt System

Int Source	Vector	Mode	Req	Ena	Pri
P3.2/int0	0003H	0	0	0	0
Timer 0	000BH	0	0	0	0
P3.3/int1	0013H	0	0	0	0
Timer 1	001BH	1	0	0	0
Serial Rcv	0023H	0	1	0	0
Serial Xmt	002BH	0	1	0	0

Selected Interrupt: EA

IT0 IEO EX0 Pri: 0

Command

Running with Code Size Limit: 2K

Load "C:\\Users\\yadal\\OneDrive\\Documents\\embedded\\Objects\\exp 17"

Call Stack - Locals

Name Location/Value Type

Call Stack - Locals Memory 1

Simulation

11: 69.81049033 sec L15 C7 CAP NUM SCR DVR R/W

27°C Partly sunny

Search

ENG IN 18:00 10-08-2026